

Show that if X is a finite complex, these maps eventually become isomorphisms. [Use induction on the number of cells of X and the five-lemma.]

3. Show that for any sequence $Z_1 \rightarrow Z_2 \rightarrow \cdots$, the natural map $\varinjlim \Omega Z_n \rightarrow \Omega \varinjlim Z_n$ is a weak homotopy equivalence, where the direct limits mean mapping telescopes.

4.G Gluing Constructions

It is a common practice in algebraic topology to glue spaces together to form more complicated spaces. In this section we describe two general procedures for making such constructions. The first is fairly straightforward but also rather rigid, lacking some homotopy invariance properties an algebraic topologist would like to see. The second type of gluing construction avoids these drawbacks by systematic use of mapping cylinders. We have already seen many special cases of both types of constructions, and having a general framework covering all these special cases should provide some conceptual clarity.

A **diagram of spaces** consists of an oriented graph Γ with a space X_v for each vertex v of Γ and a map $f_e: X_v \rightarrow X_w$ for each edge e of Γ from a vertex v to a vertex w , the words ‘from’ and ‘to’ referring to the given orientation of e . Commutativity of the diagram is not assumed. Denoting such a diagram of spaces simply by X , we define a space $\sqcup X$ to be the quotient of the disjoint union of all the spaces X_v associated to vertices of Γ under the identifications $x \sim f_e(x)$ for all maps f_e associated to edges of Γ . To give a name to this construction, let us call $\sqcup X$ the *amalgamation* of the diagram X . Here are some examples:

- If the diagram of spaces has the simple form $X_0 \xleftarrow{f} A \hookrightarrow X_1$ then $\sqcup X$ is the space $X_0 \sqcup_f X_1$ obtained from X_0 by attaching X_1 along A via f .
- A sequence of inclusions $X_1 \hookrightarrow X_2 \hookrightarrow \cdots$ determines a diagram of spaces X for which $\sqcup X$ is $\bigcup_i X_i$ with the weak topology. This holds more generally when the spaces X_i are indexed by any directed set.
- From a cover $\mathcal{U} = \{X_i\}$ of a space X by subspaces X_i we can form a diagram of spaces $X_{\mathcal{U}}$ whose vertices are the nonempty finite intersections $X_{i_1} \cap \cdots \cap X_{i_n}$ with distinct indices i_j , and whose edges are the various inclusions obtained by omitting some of the subspaces in such an intersection, for example the inclusions $X_i \cap X_j \hookrightarrow X_i$. Then $\sqcup X_{\mathcal{U}}$ equals X as a set, though possibly with a different topology. If the cover is an open cover, or if X is a CW complex and the X_i ’s are subcomplexes, then the topology will be the original topology on X .
- An action of a group G on a space X determines a diagram of spaces X_G , with X itself as the only space and with maps the homeomorphisms $g: X \rightarrow X$, $g \in G$, given by the action. In this case $\sqcup X_G$ is the orbit space X/G .

- A Δ -complex X can be viewed as a diagram of spaces X_Δ where each simplex of X gives a vertex space X_v which is a simplex of the same dimension, and the edge maps are the inclusions of faces into the simplices that contain them. Then $\sqcup X_\Delta = X$.

It can very easily happen that for a diagram of spaces X the amalgamation $\sqcup X$ is rather useless because so much collapsing has occurred that little of the original diagram remains. For example, consider a diagram X of the form $X_0 \leftarrow X_0 \times X_1 \rightarrow X_1$ whose maps are the projections onto the two factors. In this case $\sqcup X$ is simply a point. To correct for problems like this, and to get a notion with nicer homotopy-theoretic properties, we introduce the homotopy version of $\sqcup X$, which we shall denote ΔX and call the *realization* of X . Here we again start with the disjoint union of all the vertex spaces X_v , but instead of passing to a quotient space of this disjoint union, we enlarge it by filling in a mapping cylinder M_f for each map f of the diagram, identifying the two ends of this cylinder with the appropriate X_v 's. In the case of the projection diagram $X_0 \leftarrow X_0 \times X_1 \rightarrow X_1$, the union of the two mapping cylinders is the same as the quotient of $X_0 \times X_1 \times I$ with $X_0 \times X_1 \times \{0\}$ collapsed to X_0 and $X_0 \times X_1 \times \{1\}$ collapsed to X_1 . Thus ΔX is the join $X_0 * X_1$ defined in Chapter 0.

We have seen a number of other special cases of the construction ΔX . For a diagram consisting of just one map $f: X_0 \rightarrow X_1$ one gets of course the mapping cylinder M_f itself. For a diagram $X_0 \xleftarrow{f} X_1 \xrightarrow{g} X_2$ the realization ΔX is a double mapping cylinder. In case X_2 is a point this is the mapping cone of f . When the diagram has just one space and one map from this space to itself, then ΔX is the mapping torus. For a diagram consisting of two maps $f, g: X_0 \rightarrow X_1$ the space ΔX was studied in Example 2.48. Mapping telescopes are the case of a sequence of maps $X_0 \rightarrow X_1 \rightarrow \dots$. In §1.B we considered general diagrams in which the spaces are $K(G, 1)$'s.

There is a natural generalization of ΔX in which one starts with a Δ -complex Γ and a diagram of spaces associated to the 1-skeleton of Γ such that the maps corresponding to the edges of each n -simplex of Γ , $n > 1$, form a commutative diagram. We call this data a **complex of spaces**. If X is a complex of spaces, then for each n -simplex of Γ we have a sequence of maps $X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \dots \xrightarrow{f_n} X_n$, and we define the *iterated mapping cylinder* $M(f_1, \dots, f_n)$ to be the usual mapping cylinder for $n = 1$, and inductively for $n > 1$, the mapping cylinder of the composition $M(f_1, \dots, f_{n-1}) \rightarrow X_{n-1} \xrightarrow{f_n} X_n$ where the first map is the canonical projection of a mapping cylinder onto its target end. There is a natural projection $M(f_1, \dots, f_n) \rightarrow \Delta^n$, and over each face of Δ^n one has the iterated mapping cylinder for the maps associated to the edges in this face. For example when $n = 2$ one has the three mapping cylinders $M(f_1)$, $M(f_2)$, and $M(f_2 f_1)$ over the three edges of Δ^2 . All these iterated mapping cylinders over the various simplices of Γ thus fit together to form a space ΔX with a canonical projection $\Delta X \rightarrow \Gamma$. We again call ΔX the realization of the complex of spaces X , and we call Γ the *base* of X or ΔX .

Some of our earlier examples of diagrams of spaces can be regarded in a natural way as complexes of spaces:

- For a cover $\mathcal{U} = \{X_i\}$ of a space X the diagram of spaces $X_{\mathcal{U}}$ whose vertices are the finite intersections of X_i 's and whose edges are inclusions is a complex of spaces with n -simplices the n -fold inclusions. The base Γ for this complex of spaces is the barycentric subdivision of the nerve of the cover. Recall from the end of §3.3 that the nerve of a cover is the simplicial complex with n -simplices the nonempty $(n+1)$ -fold intersections of sets in the cover.
- The diagram of spaces X_G associated to an action of a group G on a space X is a complex of spaces, with n -simplices corresponding to the n -fold compositions $X \xrightarrow{g_1} X \xrightarrow{g_2} \dots \xrightarrow{g_n} X$. The base Δ -complex Γ is the $K(G, 1)$ called BG in §1.B. This was the orbit space of a free action of G on a contractible Δ -complex EG . Checking through the definitions, one sees that the space ΔX_G in this case can be regarded as the quotient of $X \times EG$ under the diagonal action of G , $g(x, y) = (g(x), g(y))$. This is the space we called the Borel construction in §3.G, with the notation $X \times_G EG$.

By a map $f: X \rightarrow Y$ of complexes of spaces over the same base Γ we mean a collection of maps $f_v: X_v \rightarrow Y_v$ for all the vertices of Γ , with commutative squares over all edges of Γ . There is then an induced map $\Delta f: \Delta X \rightarrow \Delta Y$.

Proposition 4G.1. *If all the maps f_v making up a map of complexes of spaces $f: X \rightarrow Y$ are homotopy equivalences, then so is the map $\Delta f: \Delta X \rightarrow \Delta Y$.*

Proof: The mapping cylinders $M(f_v)$ form a complex of spaces $M(f)$ over the same base Γ , and the space $\Delta M(f)$ is the mapping cylinder $M(\Delta f)$. This deformation retracts onto ΔY , so it will suffice to show that it also deformation retracts onto ΔX .

Let $M^n(\Delta f)$ be the part of $M(\Delta f)$ lying over the n -skeleton of Γ . We claim that $M^n(\Delta f) \cup \Delta X$ deformation retracts onto $M^{n-1}(\Delta f) \cup \Delta X$. It is enough to show this when $\Gamma = \Delta^n$. In this case f is a map from $X_0 \rightarrow \dots \rightarrow X_n$ to $Y_0 \rightarrow \dots \rightarrow Y_n$. By Corollary 0.20 it suffices to show that the inclusion $M^{n-1}(\Delta f) \cup \Delta X \hookrightarrow M(\Delta f)$ is a homotopy equivalence and the pair $(M(\Delta f), M^{n-1}(\Delta f) \cup \Delta X)$ satisfies the homotopy extension property. The latter assertion is evident from Example 0.15 since a mapping cylinder neighborhood is easily constructed for this pair. For the other condition, note that by induction on the dimension of Γ we may assume that $M^{n-1}(\Delta f)$ deformation retracts onto the part of ΔX over $\partial \Delta^n$. Also, the inclusion $\Delta X \hookrightarrow M(\Delta f)$ is a homotopy equivalence since it is equivalent to the map $X_n \rightarrow Y_n$, which is a homotopy equivalence by hypothesis. So Corollary 0.20 applies, and the claim that $M^n(\Delta f) \cup \Delta X$ deformation retracts onto $M^{n-1}(\Delta f) \cup \Delta X$ is proved.

Letting n vary, the infinite concatenation of these deformation retractions in the t -intervals $[1/2^{n+1}, 1/2^n]$ gives a deformation retraction of $M(\Delta f)$ onto ΔX . $\square$

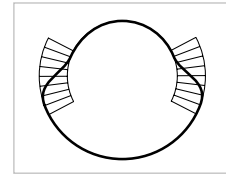
There is a canonical map $\Delta X \rightarrow \sqcup X$ induced by retracting each mapping cylinder onto its target end. In some cases this is a homotopy equivalence, for example, for a diagram $X_0 \leftarrow A \hookrightarrow X_1$ where the pair (X_1, A) has the homotopy extension property. Another example is a sequence of inclusions $X_0 \hookrightarrow X_1 \hookrightarrow \cdots$ for which the pairs (X_n, X_{n-1}) satisfy the homotopy extension property, by the argument involving mapping telescopes in the proof of Lemma 2.34. However, without some conditions on the maps it need not be true that $\Delta X \rightarrow \sqcup X$ is a homotopy equivalence, as the earlier example of the projections $X_0 \leftarrow X_0 \times X_1 \rightarrow X_1$ shows. Even with inclusion maps one need not have $\Delta X \simeq \sqcup X$ if the base Γ is not contractible. A trivial example is the diagram consisting of the two spaces Δ^0 and Δ^1 and two maps $f_0, f_1: \Delta^0 \rightarrow \Delta^1$ that happen to have the same image.

Thus one can expect the map $\Delta X \rightarrow \sqcup X$ to be a homotopy equivalence only in special circumstances. Here is one such situation:

Proposition 4G.2. *When $X_{\mathcal{U}}$ is the complex of spaces associated to an open cover $\mathcal{U} = \{X_i\}$ of a paracompact space X , the map $p: \Delta X_{\mathcal{U}} \rightarrow \sqcup X_{\mathcal{U}} = X$ is a homotopy equivalence.*

Proof: The realization $\Delta X_{\mathcal{U}}$ can also be described as the quotient space of the disjoint union of all the products $X_{i_0} \cap \cdots \cap X_{i_n} \times \Delta^n$, as the subscripts range over sets of $n+1$ distinct indices and $n \geq 0$, with the identifications over the faces of Δ^n using inclusions $X_{i_0} \cap \cdots \cap X_{i_n} \hookrightarrow X_{i_0} \cap \cdots \cap \hat{X}_{i_j} \cap \cdots \cap X_{i_n}$. From this viewpoint, points of $\Delta X_{\mathcal{U}}$ in a given ‘fiber’ $p^{-1}(x)$ can be written as finite linear combinations $\sum_i t_i x_i$ where $\sum_i t_i = 1$ and x_i is x regarded as a point of X_i , for those X_i ’s that contain x .

Since X is paracompact there is a partition of unity subordinate to the cover $\mathcal{U}$. This is a family of maps $\varphi_\alpha: X \rightarrow [0, 1]$ satisfying three conditions: The support of each φ_α is contained in some $X_{i(\alpha)}$, only finitely many φ_α ’s are nonzero near each point of X , and $\sum_\alpha \varphi_\alpha = 1$. Define a section $s: X \rightarrow \Delta X_{\mathcal{U}}$ of p by setting $s(x) = \sum_\alpha \varphi_\alpha(x) x_{i(\alpha)}$. The figure shows the case $X = S^1$ with a cover by two arcs, the heavy line indicating the image of s . In the general case the section s embeds X as a retract of $\Delta X_{\mathcal{U}}$, and it is a deformation retract since points in fibers $p^{-1}(x)$ can move linearly along line segments to $s(x)$. □



Corollary 4G.3. *If $\mathcal{U}$ is an open cover of a paracompact space X such that every nonempty intersection of finitely many sets in $\mathcal{U}$ is contractible, then X is homotopy equivalent to the nerve $N\mathcal{U}$.*

Proof: The proposition gives a homotopy equivalence $X \simeq \Delta X_{\mathcal{U}}$. Since the nonempty finite intersections of sets in $\mathcal{U}$ are contractible, the earlier proposition implies that the map $\Delta X_{\mathcal{U}} \rightarrow \Gamma$ induced by sending each intersection to a point is a homotopy equivalence. Since Γ is the barycentric subdivision of $N\mathcal{U}$, the result follows. □

Let us conclude this section with a few comments about terminology. For some diagrams of spaces such as sequences $X_1 \rightarrow X_2 \rightarrow \cdots$ the amalgamation $\sqcup X$ can be regarded as the direct limit of the vertex spaces X_v with respect to the edge maps f_e . Following this cue, the space $\sqcup X$ is commonly called the direct limit for arbitrary diagrams, even finite ones. If one views $\sqcup X$ as a direct limit, then ΔX becomes a sort of homotopy direct limit. For reasons that are explained in the next section, direct limits are often called ‘colimits.’ This has given rise to the rather unfortunate name of ‘hocolim’ for ΔX , short for ‘homotopy colimit.’ In preference to this we have chosen the term ‘realization,’ both for its intrinsic merits and because ΔX is closely related to what is called the geometric realization of a simplicial space.

Exercises

1. Show that for a sequence of maps $X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \cdots$, the infinite iterated mapping cylinder $M(f_1, f_2, \cdots)$, which is the union of the finite iterated mapping cylinders $M(f_1, \cdots, f_n)$, deformation retracts onto the mapping telescope.
2. Show that if X is a complex of spaces in which all the maps are homeomorphisms, then the projection $\Delta X \rightarrow \Gamma$ is a fiber bundle.
3. What is the nerve of the cover of a simplicial complex by the open stars of its vertices? [See Lemma 2C.2.]
4. Show that Proposition 4G.2 and its corollary hold also for CW complexes and covers by families of subcomplexes. [CW complexes are paracompact; see [VBKT].]

4.H Eckmann-Hilton Duality

There is a very nice duality principle in homotopy theory, called Eckmann-Hilton duality in its more refined and systematic aspects, but which in its most basic form involves the simple idea of reversing the direction of all arrows in a given construction. For example, if in the definition of a fibration as a map satisfying the homotopy lifting property we reverse the direction of all the arrows, we obtain the dual notion of a **cofibration**. This is a map $i: A \rightarrow B$ satisfying the following property: Given $\tilde{g}_0: B \rightarrow X$ and a homotopy $g_t: A \rightarrow X$ such that $\tilde{g}_0 i = g_0$, there exists a homotopy $\tilde{g}_t: B \rightarrow X$ such that $\tilde{g}_t i = g_t$. In the special case that i is the inclusion of a subspace, this is the homotopy extension property, and the next proposition says that this is indeed the general case. So a cofibration is the same as an inclusion satisfying the homotopy extension property.

Proposition 4H.1. *If $i: A \rightarrow B$ is a cofibration, then i is injective, and in fact a homeomorphism onto its image.*

$$\begin{array}{ccc} A & \xrightarrow{g_t} & X \\ i \downarrow & \nearrow \tilde{g}_0 & \\ B & & \end{array}$$